COMPUTER ENGINEERING

QUEZON CITY

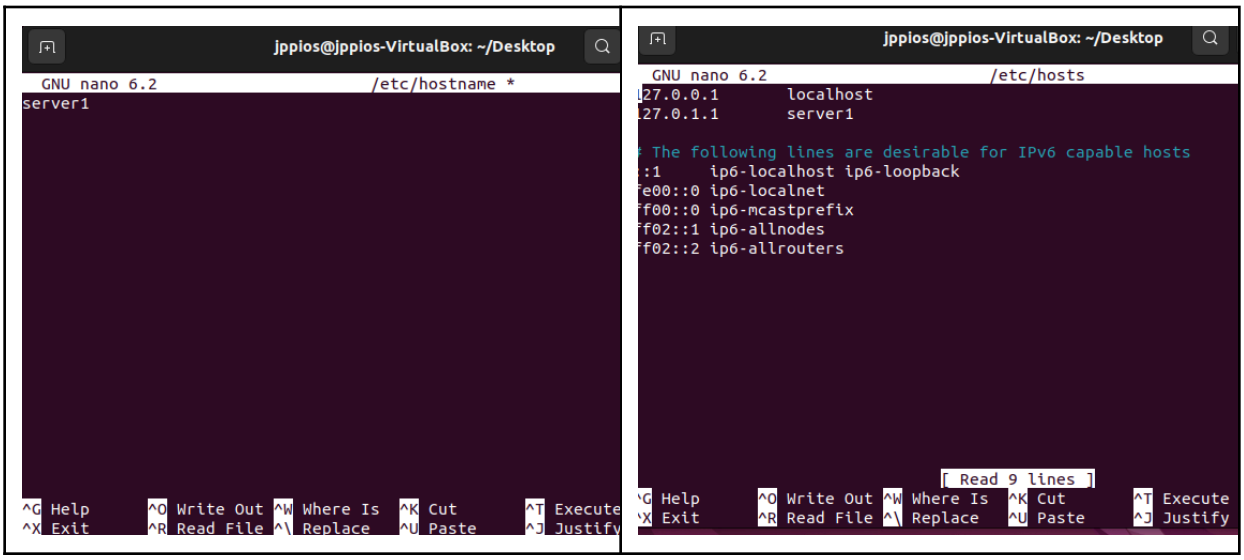
COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	July 15, 2026	**DATE SUBMITTED:**	July 15, 2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

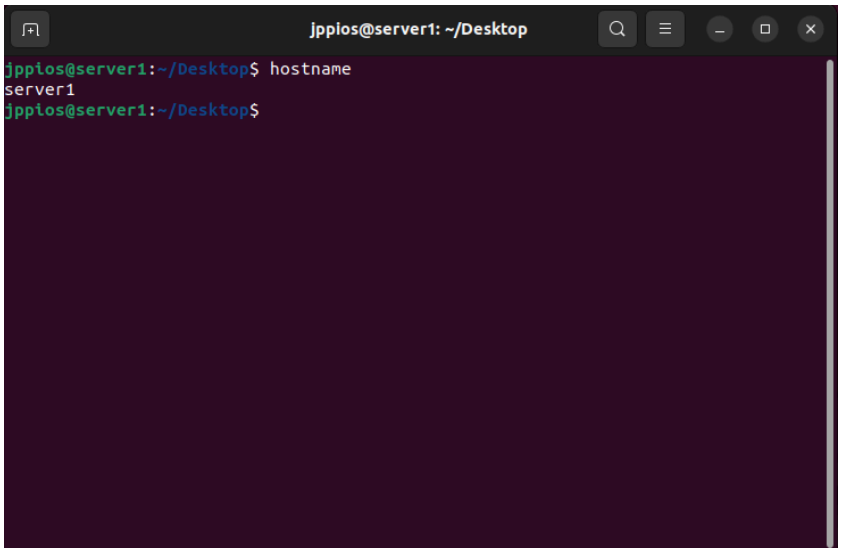
ACTIVITY NO. 1.1 Creating Virtual Machines

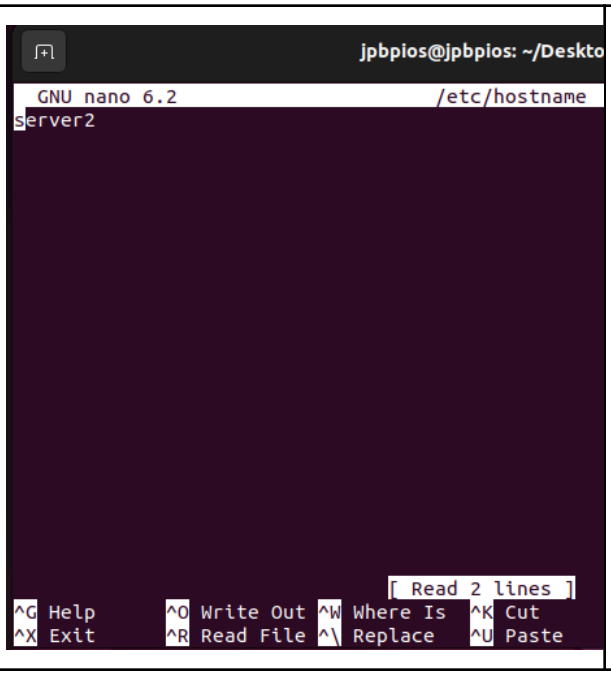
1. PROCEDURE OUTPUT

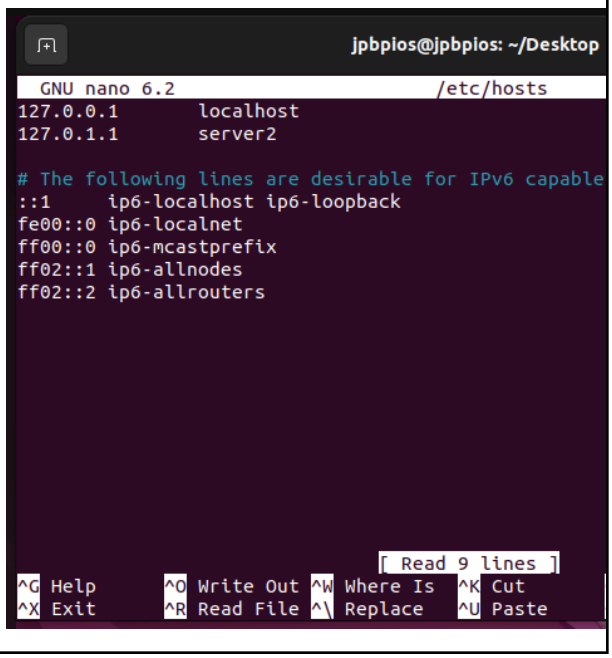
What procedures did you perform?

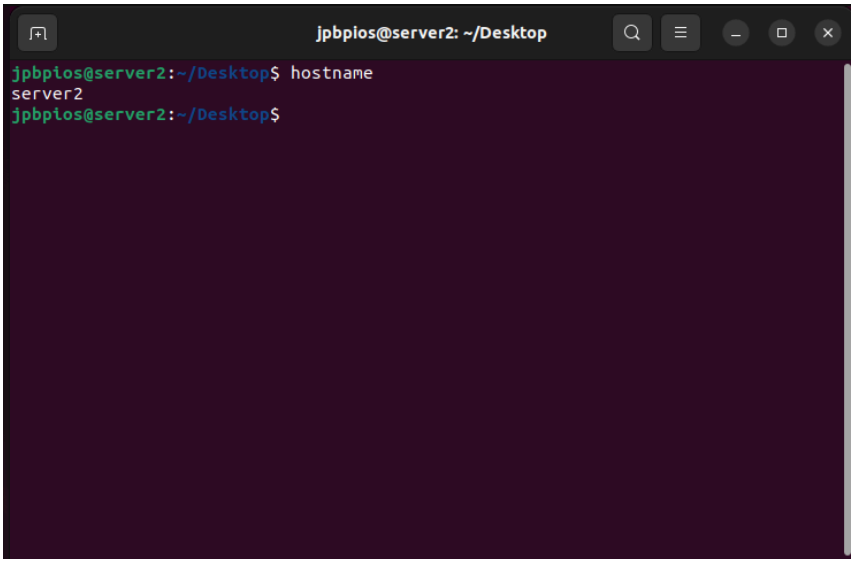
Task 1



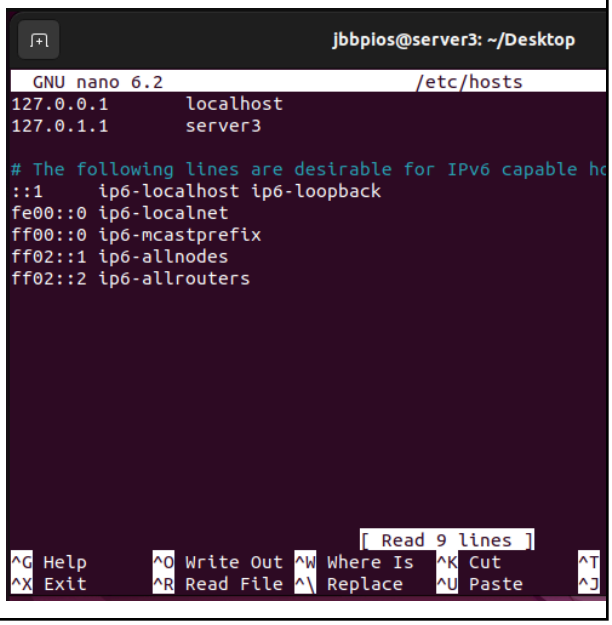

jppios@server1: ~/Desktop
jppios@server1:~/Desktop$ hostname
server1
jppios@server1:~/Desktop$

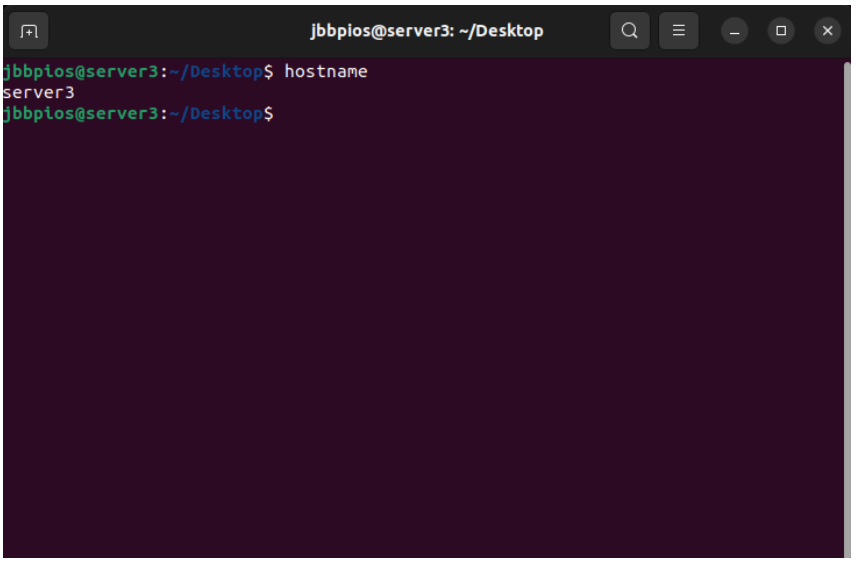

jpbpios@jpbpios: ~/Deskto
GNU nano 6.2 /etc/hostname
server2
[Read 2 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


jpbpios@jpbpios: ~/Desktop
GNU nano 6.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 server2
The following lines are desirable for IPv6 capable
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
[Read 9 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste

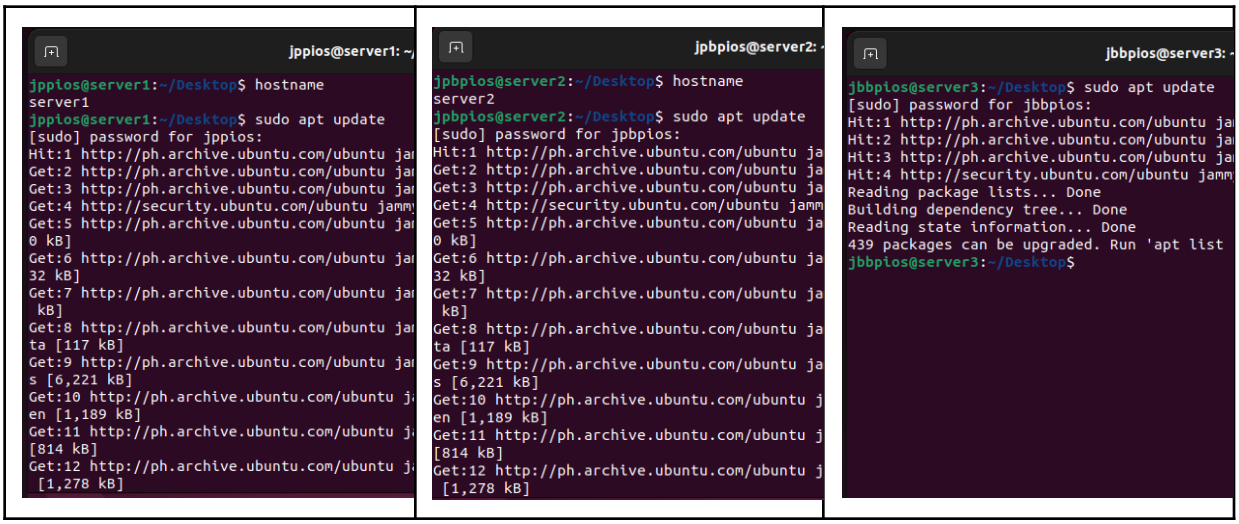

jpbpios@server2: ~/Desktop
jpbpios@server2:~/Desktop$ hostname
server2
jpbpios@server2:~/Desktop$


jbbpios@server3: ~/Desktop
GNU nano 6.2 /etc/hostname
server3
[Read 2 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T
^X Exit ^R Read File ^\ Replace ^U Paste ^J


jbbpios@server3: ~/Desktop
GNU nano 6.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 server3
The following lines are desirable for IPv6 capable ho
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
[Read 9 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T
^X Exit ^R Read File ^\ Replace ^U Paste ^J

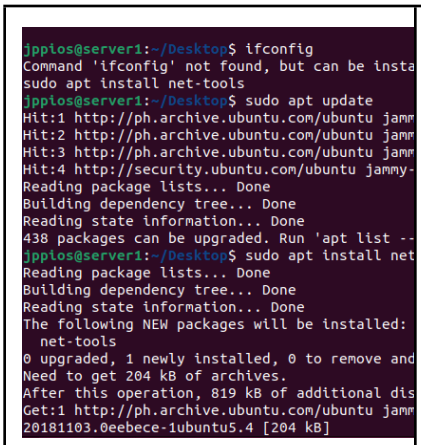

jbbpios@server3: ~/Desktop
jbbpios@server3:~/Desktop$ hostname
server3
jbbpios@server3:~/Desktop$

Task 2

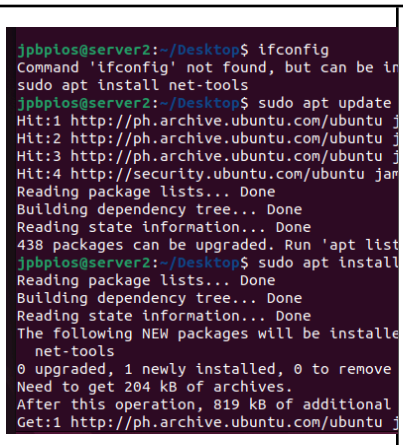

jppios@server1: ~/
jppios@server1:~/Desktop$ hostname
server1
jppios@server1:~/Desktop$ sudo apt update
[sudo] password for jppios:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jar
Get:2 http://ph.archive.ubuntu.com/ubuntu jar
Get:3 http://ph.archive.ubuntu.com/ubuntu jar
Get:4 http://security.ubuntu.com/ubuntu jammy
Get:5 http://ph.archive.ubuntu.com/ubuntu jar
0 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu jar
32 kB]
Get:7 http://ph.archive.ubuntu.com/ubuntu jar
kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu jar
ta [117 kB]
Get:9 http://ph.archive.ubuntu.com/ubuntu jar
s [6,221 kB]
Get:10 http://ph.archive.ubuntu.com/ubuntu j
en [1,189 kB]
Get:11 http://ph.archive.ubuntu.com/ubuntu j
[814 kB]
Get:12 http://ph.archive.ubuntu.com/ubuntu j
[1,278 kB]
jpbpios@server2: ~/
jpbpios@server2:~/Desktop$ hostname
server2
jpbpios@server2:~/Desktop$ sudo apt update
[sudo] password for jpbpios:
Hit:1 http://ph.archive.ubuntu.com/ubuntu ja
Get:2 http://ph.archive.ubuntu.com/ubuntu ja
Get:3 http://ph.archive.ubuntu.com/ubuntu ja
Get:4 http://security.ubuntu.com/ubuntu jamm
Get:5 http://ph.archive.ubuntu.com/ubuntu ja
0 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu ja
32 kB]
Get:7 http://ph.archive.ubuntu.com/ubuntu ja
kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu ja
ta [117 kB]
Get:9 http://ph.archive.ubuntu.com/ubuntu ja
s [6,221 kB]
Get:10 http://ph.archive.ubuntu.com/ubuntu j
en [1,189 kB]
Get:11 http://ph.archive.ubuntu.com/ubuntu j
[814 kB]
Get:12 http://ph.archive.ubuntu.com/ubuntu j
[1,278 kB]
jbbpios@server3: ~/
jbbpios@server3:~/Desktop$ sudo apt update
[sudo] password for jbbpios:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jar
Hit:2 http://ph.archive.ubuntu.com/ubuntu jar
Hit:3 http://ph.archive.ubuntu.com/ubuntu jar
Hit:4 http://security.ubuntu.com/ubuntu jammy
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
439 packages can be upgraded. Run 'apt list
jbbpios@server3:~/Desktop$


jppios@server1: ~/D
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
439 packages can be upgraded. Run 'apt list --
jppios@server1:~/Desktop$ sudo apt install ope
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be inst
ncurses-term openssh-client openssh-sftp-ser
Suggested packages:
keychain libpam-ssh monkeysphere ssh-askpass
The following NEW packages will be installed:
ncurses-term openssh-server openssh-sftp-ser
The following packages will be upgraded:
openssh-client
1 upgraded, 4 newly installed, 0 to remove and
Need to get 1,656 kB of archives.
After this operation, 6,056 kB of additional
Get:1 http://ph.archive.ubuntu.com/ubuntu jamm
t amd64 1:8.9p1-3ubuntu0.16 [904 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jamm
jppios@server1:~/Desktop$ sudo service
jppios@server1:~/Desktop$ sudo systemctl
ssh.service - OpenBSD Secure Shell ser
Loaded: loaded (/lib/systemd/syster
Active: active (running) since Wed
Docs: man:sshd(8)
man:sshd_config(5)
Main PID: 3189 (sshd)
Tasks: 1 (limit: 4606)
Memory: 1.7M
CPU: 41ms
CGroup: /system.slice/ssh.service
└─3189 "sshd: /usr/sbin/ssh
Jul 15 22:23:16 server1 systemd[1]: Star
Jul 15 22:23:16 server1 sshd[3189]: Serv
Jul 15 22:23:16 server1 sshd[3189]: Serv
Jul 15 22:23:16 server1 systemd[1]: Star
lines 1-16/16 (END)
jppios@server1:~/Desktop$ sudo ufw al
Rules updated
Rules updated (v6)
jppios@server1:~/Desktop$ sudo ufw en
Firewall is active and enabled on sys
jppios@server1:~/Desktop$ sudo ufw st
Status: active
To Action
-- ------
22/tcp ALLOW
22/tcp (v6) ALLOW
jppios@server1:~/Desktop$
jpbpios@server2: ~/
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
439 packages can be upgraded. Run 'apt list --
jpbpios@server2:~/Desktop$ sudo apt install op
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be inst
ncurses-term openssh-client openssh-sftp-ser
Suggested packages:
keychain libpam-ssh monkeysphere ssh-askpass
The following NEW packages will be installed:
ncurses-term openssh-server openssh-sftp-ser
The following packages will be upgraded:
openssh-client
1 upgraded, 4 newly installed, 0 to remove and
Need to get 1,656 kB of archives.
After this operation, 6,056 kB of additional
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jamm
t amd64 1:8.9p1-3ubuntu0.16 [904 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jamm
server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Processing triggers for ufw (0.36.1-4ubu
jpbpios@server2:~/Desktop$ sudo service
jpbpios@server2:~/Desktop$ sudo systemct
ssh.service - OpenBSD Secure Shell ser
Loaded: loaded (/lib/systemd/syster
Active: active (running) since Wed
Docs: man:sshd(8)
man:sshd_config(5)
Main PID: 2957 (sshd)
Tasks: 1 (limit: 4606)
Memory: 1.7M
CPU: 37ms
CGroup: /system.slice/ssh.service
└─2957 "sshd: /usr/sbin/ssh
Jul 15 22:24:32 server2 systemd[1]: Star
Jul 15 22:24:32 server2 sshd[2957]: Serv
Jul 15 22:24:32 server2 sshd[2957]: Serv
Jul 15 22:24:32 server2 systemd[1]: Star
lines 1-16/16 (END)
jpbpios@server2:~/Desktop$ sudo ufw al
Rules updated
Rules updated (v6)
jpbpios@server2:~/Desktop$ sudo ufw en
Firewall is active and enabled on syst
jpbpios@server2:~/Desktop$ sudo ufw st
Status: active
To Action
-- ------
22/tcp ALLOW
22/tcp (v6) ALLOW
jpbpios@server2:~/Desktop$
jbbpios@server3: ~/De
439 packages can be upgraded. Run 'apt list --u
jbbpios@server3:~/Desktop$ sudo apt install ope
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be insta
ncurses-term openssh-client openssh-sftp-serv
Suggested packages:
keychain libpam-ssh monkeysphere ssh-askpass
The following NEW packages will be installed:
ncurses-term openssh-server openssh-sftp-serv
The following packages will be upgraded:
openssh-client
1 upgraded, 4 newly installed, 0 to remove and
Need to get 1,656 kB of archives.
After this operation, 6,056 kB of additional di
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy
t amd64 1:8.9p1-3ubuntu0.16 [904 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy
server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy
r amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy
jbbpios@server3:~/Desktop$ sudo servi
jbbpios@server3:~/Desktop$ sudo syste
ssh.service - OpenBSD Secure Shell
Loaded: loaded (/lib/systemd/sys
Active: active (running) since W
Docs: man:sshd(8)
man:sshd_config(5)
Main PID: 2619 (sshd)
Tasks: 1 (limit: 4606)
Memory: 1.7M
CPU: 25ms
CGroup: /system.slice/ssh.servic
└─2619 "sshd: /usr/sbin/
Jul 15 22:40:18 server3 systemd[1]: S
Jul 15 22:40:18 server3 sshd[2619]: S
Jul 15 22:40:18 server3 sshd[2619]: S
Jul 15 22:40:18 server3 systemd[1]: S
jbbpios@server3:~/Desktop$ sudo ufw al
Rules updated
Rules updated (v6)
jbbpios@server3:~/Desktop$ sudo ufw en
Firewall is active and enabled on syst
jbbpios@server3:~/Desktop$ sudo ufw st
Status: active
To Action
-- ------
22/tcp ALLOW
22/tcp (v6) ALLOW
jbbpios@server3:~/Desktop$

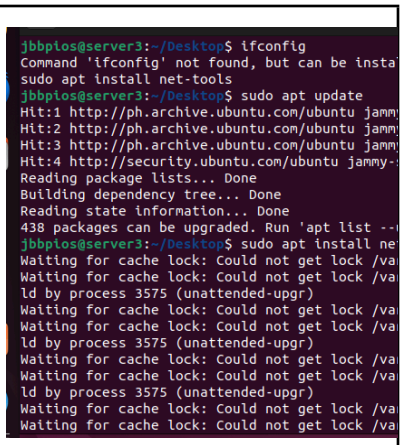
Task 3



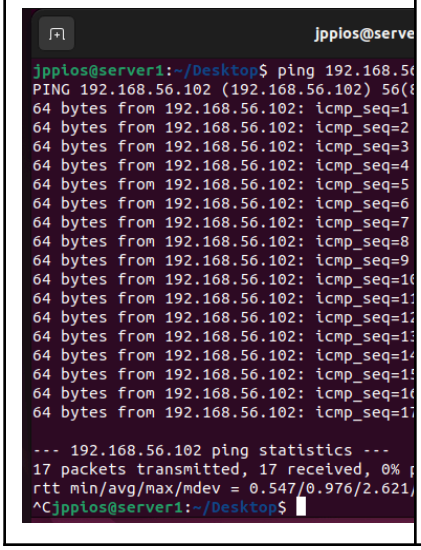
10.0.2.15
192.168.56.101

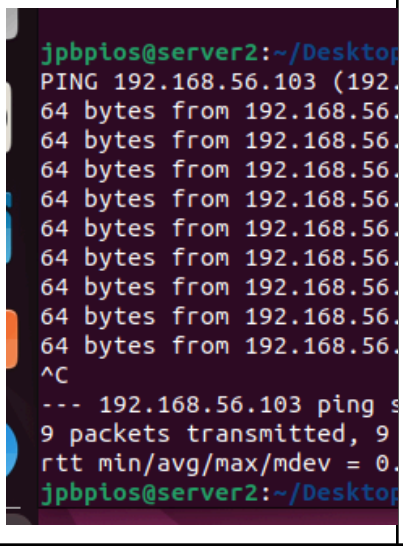


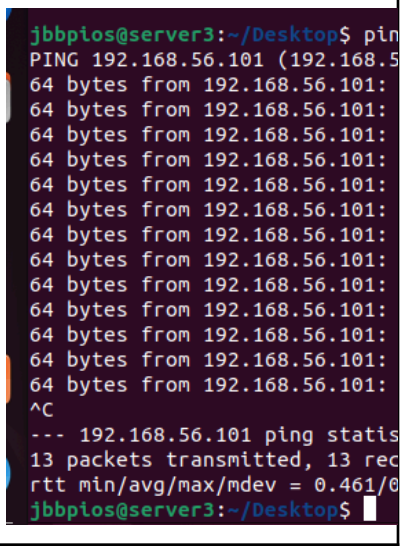
10.0.2.15
192.168.56.102



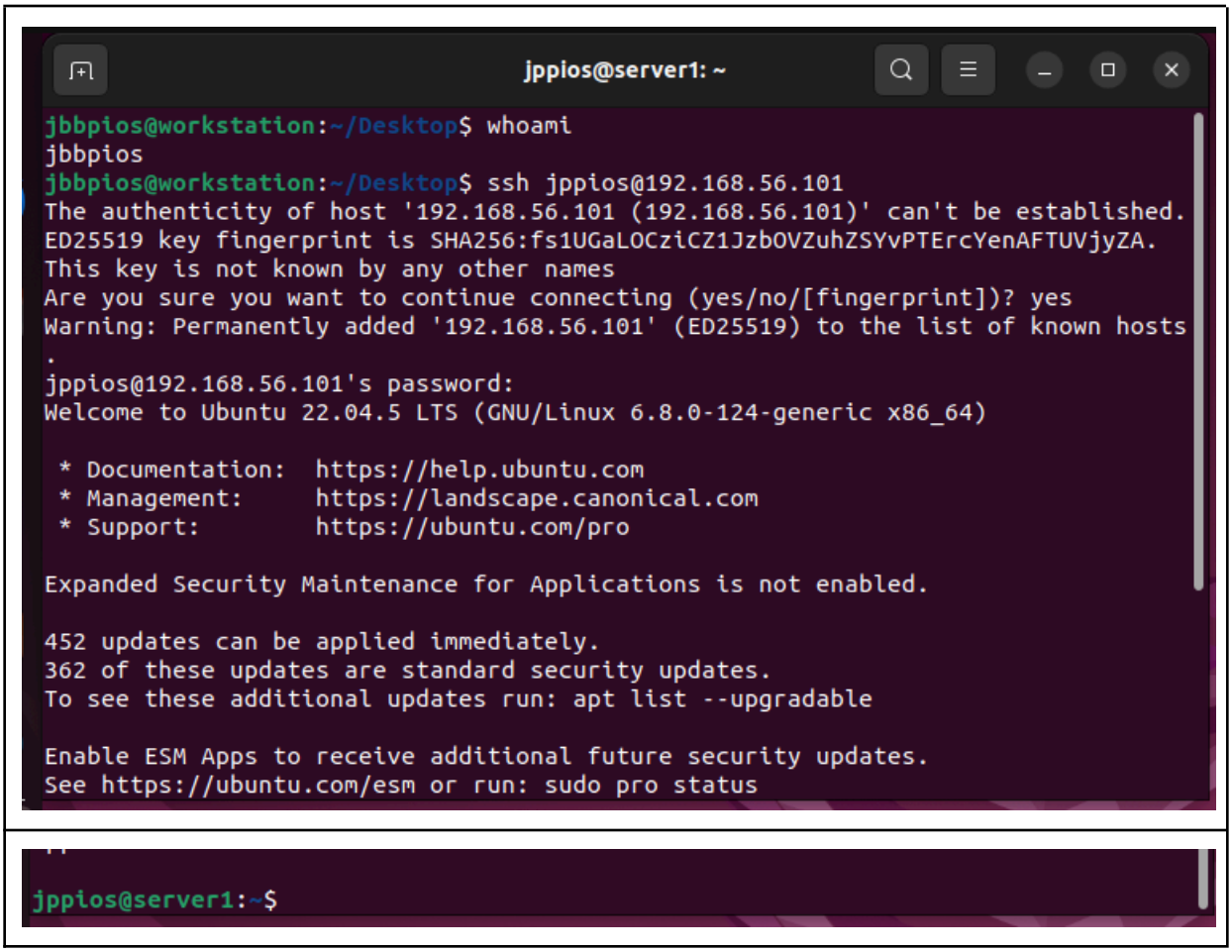
10.0.2.15
192.168.56.103

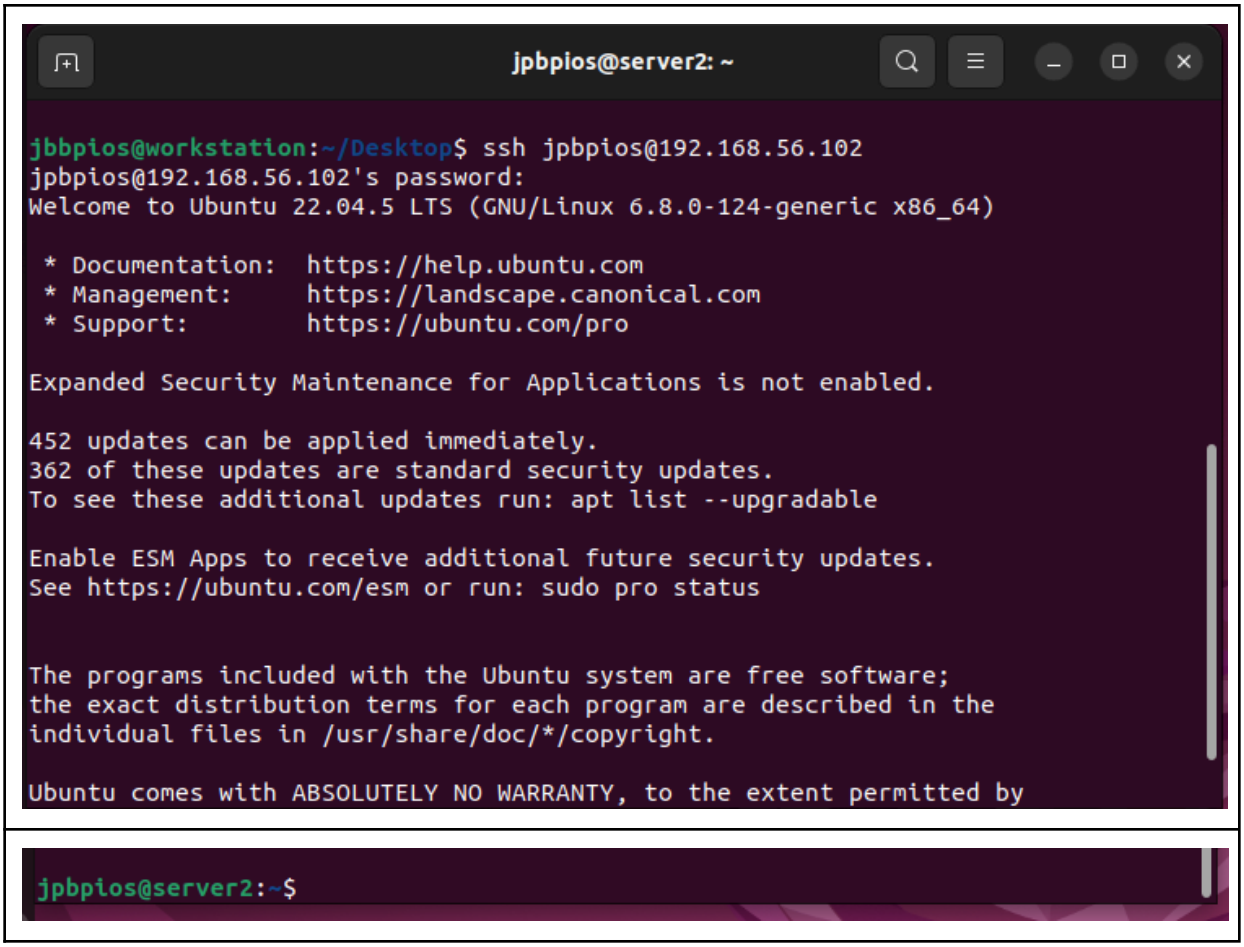




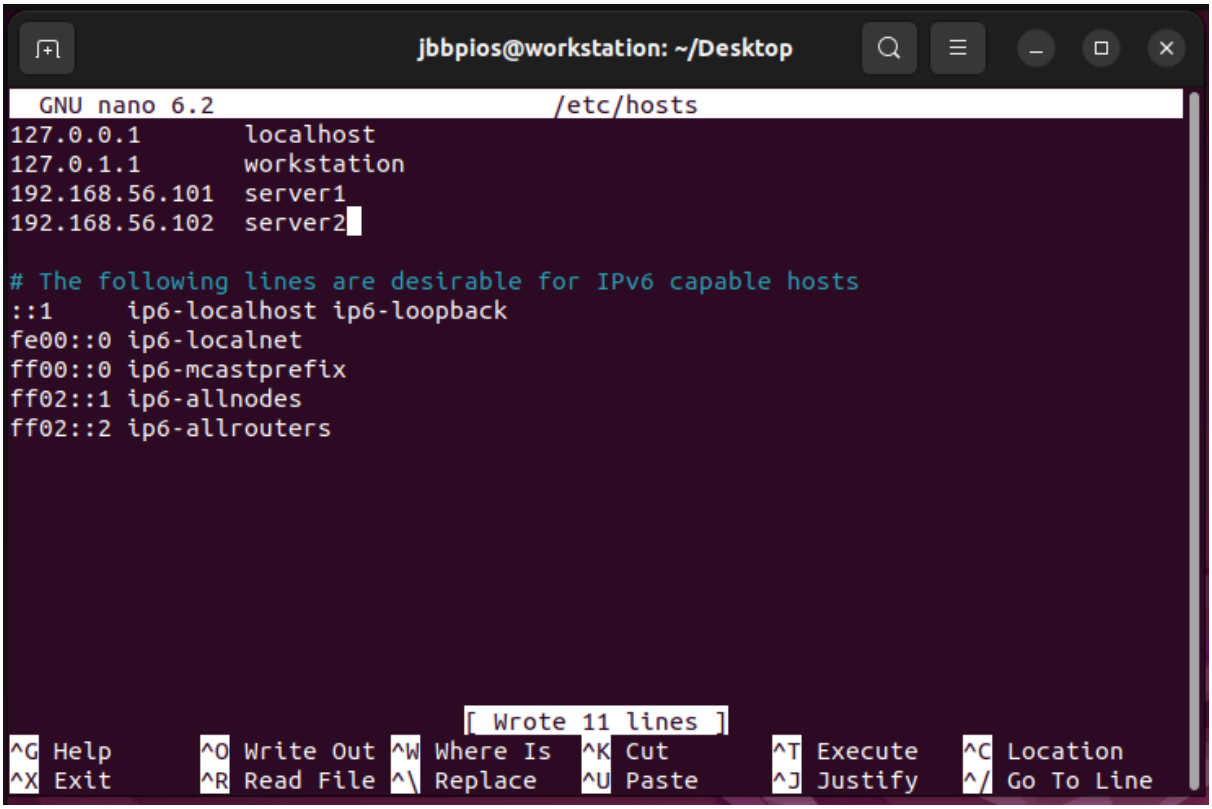


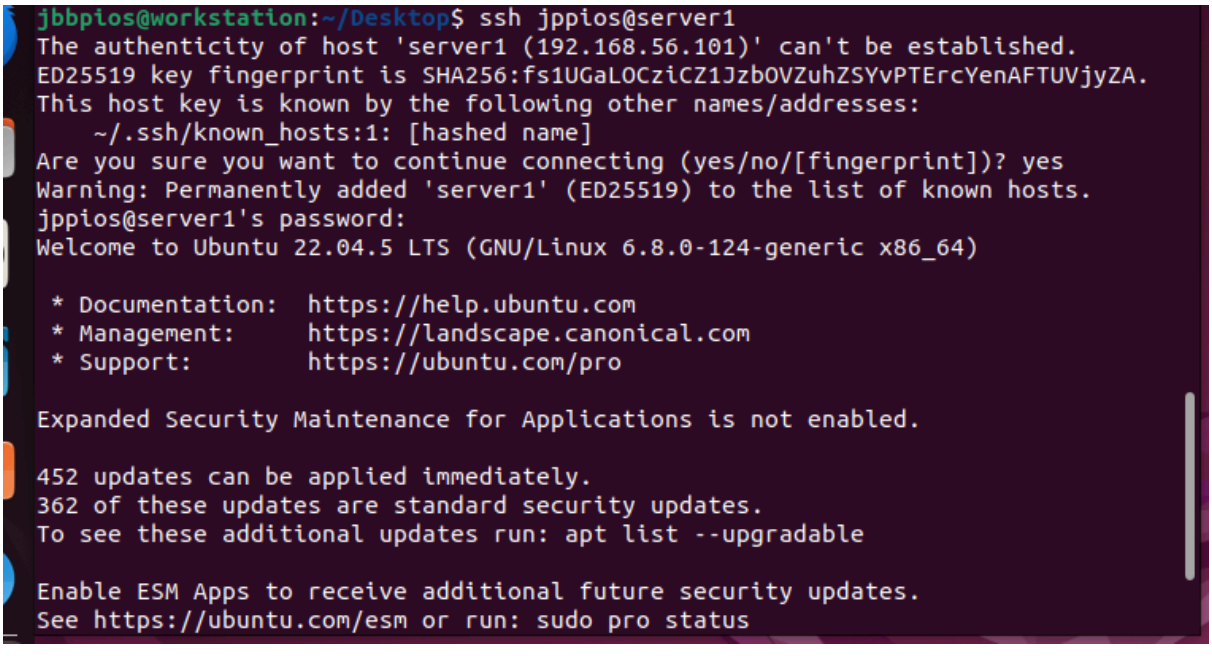
Task 4


jppios@server1: ~
jbbpios@workstation:~/Desktop$ whoami
jbbpios
jbbpios@workstation:~/Desktop$ ssh jppios@192.168.56.101
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:fs1UGaLOCziCZ1JzbOVZuhZSYvPTErcYenAFTUVjyZA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts
.
jppios@192.168.56.101's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
452 updates can be applied immediately.
362 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
jppios@server1:~$


jpbpios@server2: ~
jbbpios@workstation:~/Desktop$ ssh jpbpios@192.168.56.102
jpbpios@192.168.56.102's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
452 updates can be applied immediately.
362 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by

jpbpios@server2:~$


jbbpios@workstation: ~/Desktop
GNU nano 6.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 workstation
192.168.56.101 server1
192.168.56.102 server2
The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
[Wrote 11 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


jbbpios@workstation:~/Desktop$ ssh jppios@server1
The authenticity of host 'server1 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:fs1UGaLOCziCZ1JzbOVZuhZSYvPTErcYenAFTUVjyZA.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
jppios@server1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
452 updates can be applied immediately.
362 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```
jbbpios@workstation:~/Desktop$ ssh jpbpios@server2
The authenticity of host 'server2 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:Dw40znLVc9CdFEnWNlYNNEXI1FB26PieqVAbse8OdMI.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
jpbpios@server2's password:
Permission denied, please try again.
jpbpios@server2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   - We can use the hostname instead of the IP address because the /etc/hosts file maps hostnames to their corresponding IP addresses. When an SSH command uses a hostname, the operating system looks it up in the hosts file and automatically connects to the correct IP address.

2. How secured is SSH?
   - SSH is a highly secure protocol because it encrypts all data transmitted between the client and the server, preventing unauthorized users from intercepting sensitive information. It also supports secure authentication methods, such as passwords and SSH keys, making remote access much safer than unsecured protocols.

COMPUTER ENGINEERING

QUEZON CITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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